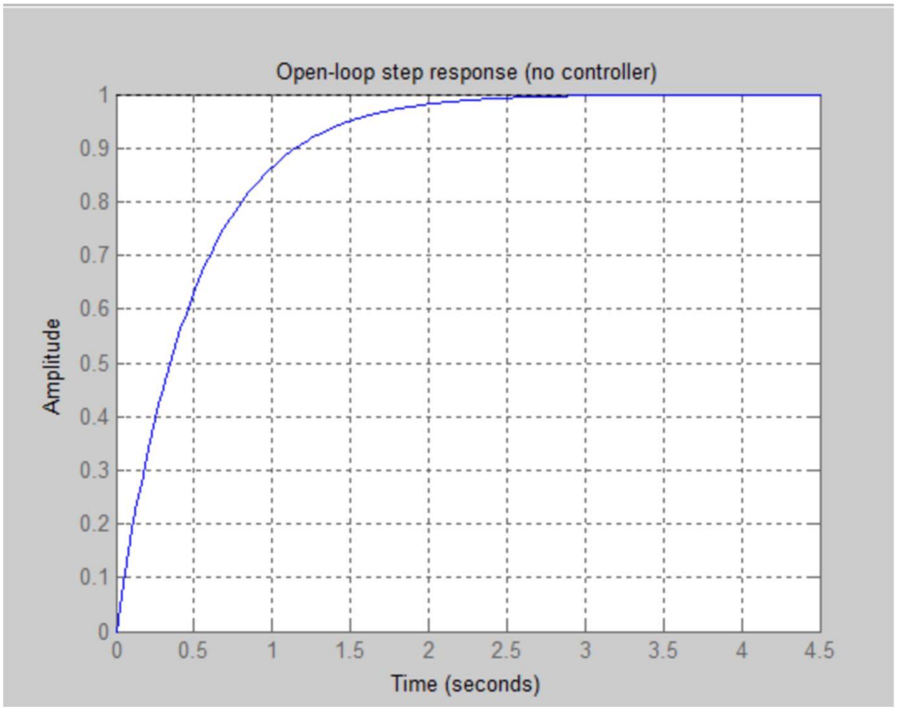
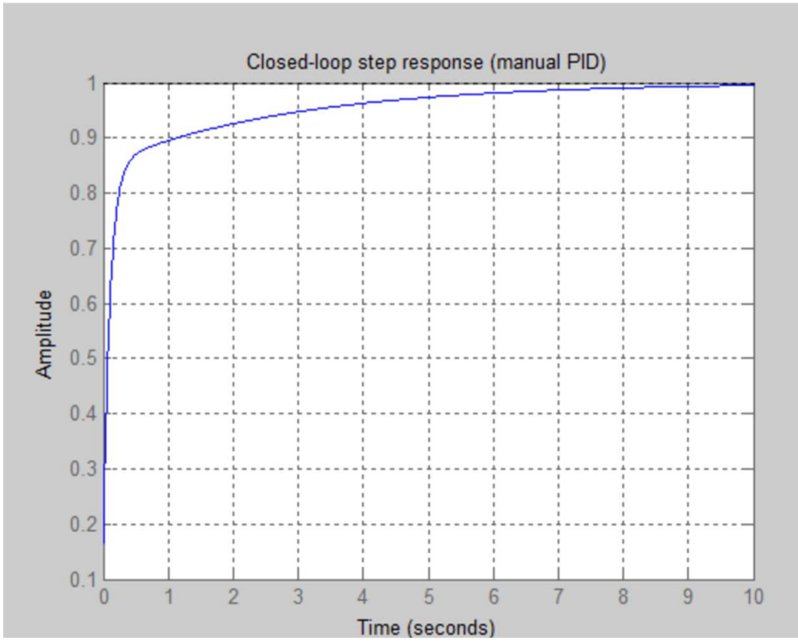
<u>PID controller (DC motor speed control) figures:</u>

Figure 1:



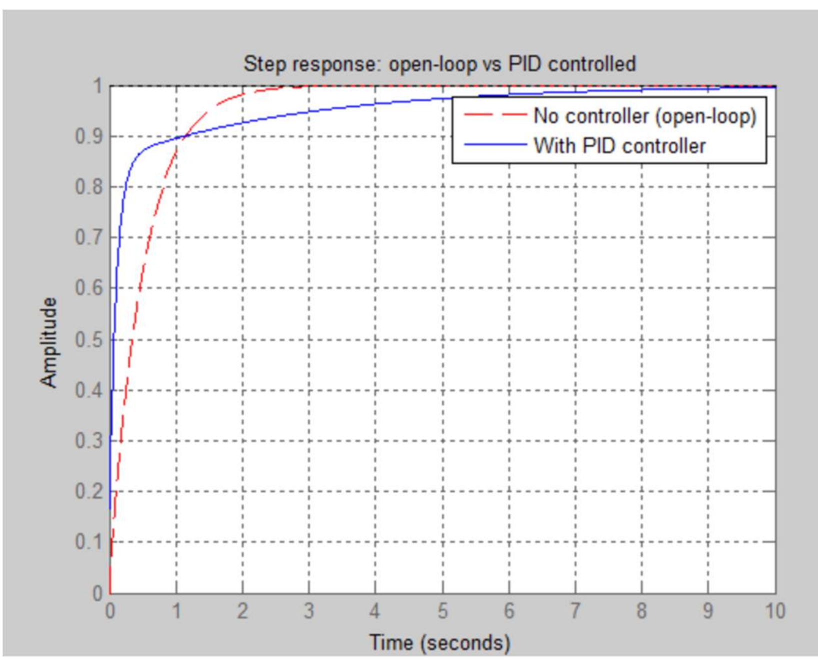
Open-loop step response of the DC motor with no controller. The system rises slowly, taking around 2.5 seconds to settle at steady-state amplitude of 1, showing the sluggish uncontrolled behaviour of the plant.

Figure 2:



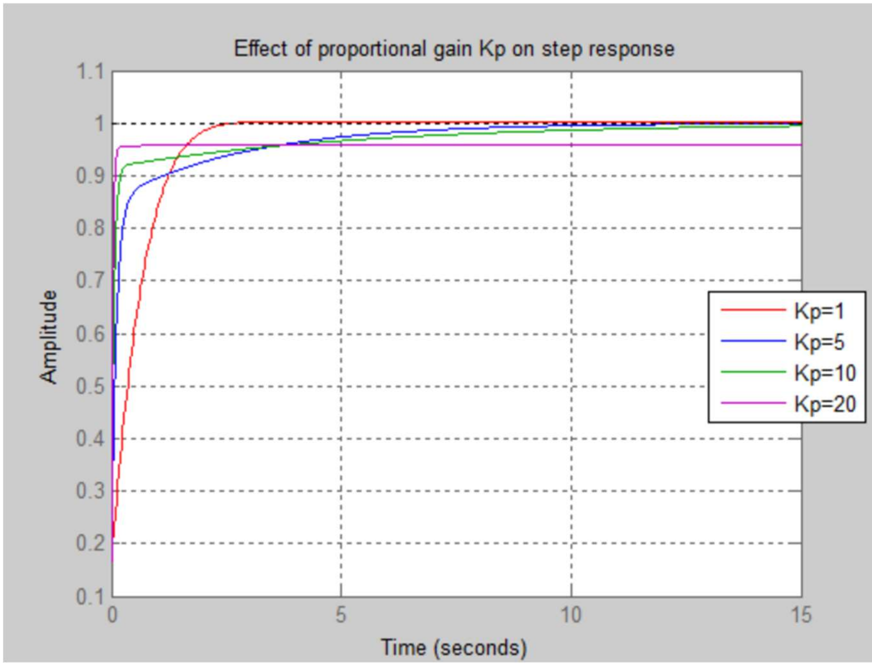
Closed-loop step response after applying the manually tuned PID controller. The system reaches steady-state significantly faster with no overshoot, settling cleanly at amplitude 1.

Figure 3:



Direct comparison of open-loop (red dashed) vs PID-controlled (blue solid) step responses. The PID controller achieves a faster initial rise while the open-loop response, despite settling earlier, lacks the precision and tunability of the controlled system.

Figure 4:



Effect of varying proportional gain Kp (1, 5, 10, 20) on step response. Higher Kp values increase rise speed but introduce steady-state error at Kp=20 (magenta curve plateaus below 1), demonstrating the trade-off between speed and accuracy in PID tuning.